

# Variational Analysis/Numerical Optimization PhD Qualifier – January 2026

Do 8 problems, 4 from problems 1–5 and 4 from problems 6–10.

1. Recall that  $f : \mathbb{R}^n \rightarrow \mathbb{R}$  is convex if and only if

$$f[(1 - \alpha)\mathbf{x} + \alpha\mathbf{y}] \leq (1 - \alpha)f(\mathbf{x}) + \alpha f(\mathbf{y})$$

for all  $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$  and  $\alpha \in [0, 1]$ .

- (a) If  $f$  is continuously differentiable, then show that  $f$  is convex if and only if

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})(\mathbf{y} - \mathbf{x}) \quad \text{for all } \mathbf{x}, \mathbf{y} \in \mathbb{R}^n.$$

- (b) If  $f$  is twice continuously differentiable, then show that  $f$  is convex if and only if  $\nabla^2 f$  is positive semidefinite on  $\mathbb{R}^n$ .

2. Given  $\mathbf{x}_0 \in \mathbb{R}^n$ , let  $0 < \sigma < \mu < 1$  be Wolfe line search parameters. In a Wolfe line search for minimizing  $f : \mathbb{R}^n \rightarrow \mathbb{R}$ ,  $\mathbf{x}_{k+1} = \mathbf{x}_k - s_k \mathbf{g}_k$  where  $\mathbf{g}_k = \nabla f(\mathbf{x}_k)^T$  and the stepsize  $s_k$  at step  $k$  is chosen so that the following two conditions hold:

**W1.**  $F(s_k) \leq F_L(s_k)$ , where  $F(s) = f(\mathbf{x}_k - s\mathbf{g}_k)$  and  $F_L(s) = f(\mathbf{x}_k) - \sigma s \|\mathbf{g}_k\|^2$ .

**W2.**  $F'(s_k) \geq \mu F'(0) = -\mu \|\mathbf{g}_k\|^2$ .

If  $f$  is bounded from below, prove that there exists a stepsize  $s_k > 0$  satisfying the Wolfe conditions. If  $f$  is twice continuously differentiable over the convex hull of the level set

$$L = \{\mathbf{x} \in \mathbb{R}^n : f(\mathbf{x}) \leq f(\mathbf{x}_0)\},$$

and the largest eigenvalue of  $\nabla^2 f$  is bounded by  $M < \infty$  over  $L$ , then prove that the gradient  $\mathbf{g}_k$  converges to zero.

3. Suppose  $f : \mathcal{K} \rightarrow \mathbb{R}$  where  $\mathcal{K} \subset \mathbb{R}^n$  is a convex set.

- (a) If  $\mathbf{x}^*$  is a local minimizer of  $f$ , then show that

$$\nabla f(\mathbf{x}^*)(\mathbf{x} - \mathbf{x}^*) \geq 0 \quad \text{for all } \mathbf{x} \in \mathcal{K}. \tag{1}$$

- (b) If  $f$  is convex over  $\mathcal{K}$  and differentiable at some point  $\mathbf{x}^* \in \mathcal{K}$  where (1) holds, then show that  $\mathbf{x}^*$  is a global minimizer for  $f$  over  $\mathcal{K}$ .

- (c) Consider the quadratic programming problem

$$\min \frac{1}{2} \mathbf{x}^T \mathbf{A} \mathbf{x} - \mathbf{b}^T \mathbf{x} \quad \text{subject to } \mathbf{x} \in \mathcal{K},$$

where  $\mathcal{K} \subset \mathbb{R}^n$  is a closed convex set and  $\mathbf{A} \in \mathbb{R}^{n \times n}$  is positive definite with smallest eigenvalue  $\alpha > 0$ . Show that the solutions  $\mathbf{u}_i$ ,  $i = 1, 2$ , associated with  $\mathbf{b} = \mathbf{b}_i$ ,  $i = 1, 2$  respectively, satisfy

$$\|\mathbf{u}_1 - \mathbf{u}_2\| \leq \|\mathbf{b}_1 - \mathbf{b}_2\| / \alpha.$$

4. Prove the arithmetic-geometric mean inequality:

$$\left( \prod_{i=1}^n x_i \right)^{1/n} \leq \frac{1}{n} \sum_{i=1}^n x_i$$

where  $\mathbf{x} \geq \mathbf{0}$ . *Hint:* change variables to  $\mathbf{y} = c\mathbf{x}$  where  $c$  is a scalar chosen so that the components of  $\mathbf{y}$  sum to 1. Note that maximizing the product of the  $y_i$  is equivalent to maximizing the log of the product since  $\log(t)$  is a monotone increasing function of  $t$  on  $(0, \infty)$ . Using duality, the max of the log of the product can be determined explicitly.

5. Consider the quadratic program

$$\min \frac{1}{2} \mathbf{x}^T \mathbf{A} \mathbf{x} - \mathbf{b}^T \mathbf{x} \quad \text{subject to } \mathbf{a}^T \mathbf{x} = \mathbf{c}, \quad (2)$$

where  $\mathbf{A}$  is a symmetric, positive definite matrix. The penalty approximation to (2) is

$$\min \frac{1}{2} \mathbf{x}^T \mathbf{A} \mathbf{x} - \mathbf{b}^T \mathbf{x} + \frac{p}{2} (\mathbf{a}^T \mathbf{x} - \mathbf{c})^2, \quad (3)$$

where  $p > 0$  is the penalty parameter. Use the first-order optimality conditions for both (2) and (3) to obtain a bound for the distance between the solution of (2) and the solution of (3).

6. Let us consider the minimization problem

$$\min I(y) := \int_0^1 f(y'(x), y(x), x) dx \quad \text{subject to } y \in C_0^1, \quad (4)$$

where  $f$  is a given continuously differentiable function. If  $y^*$  is a local minimizer of this problem,  $s$  is a scalar,  $h \in C_0^1$ , then setting to zero the derivative of  $I(y^* + sh)$  evaluated at  $s = 0$  yields the following first-order optimality condition:

$$\int_0^1 \left[ \left( \frac{\partial f}{\partial y'} \right)^* (x) h'(x) + \left( \frac{\partial f}{\partial y} \right)^* (x) h(x) \right] dx = 0 \quad (5)$$

for all  $h \in C_0^1$ , where

$$\left( \frac{\partial f}{\partial y'} \right)^* (x) := \frac{\partial f}{\partial y'}(y^{*'}(x), y^*(x), x) \quad \text{and} \quad \left( \frac{\partial f}{\partial y} \right)^* (x) := \frac{\partial f}{\partial y}(y^{*'}(x), y^*(x), x).$$

- (a) As a consequence of (5), show that

$$\left( \frac{\partial f}{\partial y'} \right)^* \in C^1 \quad (6)$$

and the Euler equation is satisfied. Note that it is only assumed that  $f$  is continuously differentiable; you should show that at a local minimizer of (4), the partial derivative in (6) is continuously differentiable.

(b) For the special case

$$f(y'(x), y(x), x) = \frac{1}{2}y'(x)^2 - \varphi(x)y(x)^2,$$

where  $\varphi \in C^k$  for some integer  $k \geq 0$ , what does the observation in (a) imply concerning the smoothness of a minimizer  $y^*$  for (4)?

7. Suppose that  $P$  and  $Q$  are real-valued functions defined on the interval  $[0, 1]$  with  $P$  continuously differentiable and  $Q$  continuous. It is assumed that there exists  $\alpha > 0$  such that  $P(x) \geq \alpha$  for all  $x \in [0, 1]$ . Define by

$$(h) := \int_0^1 P(x)h'(x)^2 + Q(x)h(x)^2 dx.$$

Suppose that  $(h)$  has the following property: There exists  $\gamma > 0$  such that

$$(h) \geq \int_0^1 h'(x)^2 + h(x)^2 dx \quad \text{for all } h \in H_0^1,$$

where  $H_0^1$  is the space of functions with  $h'$  square integrable over  $[0, 1]$  and  $h(0) = h(1) = 0$ .

Prove the following: If  $u$  is a solution of the boundary-value problem

$$\frac{d}{dx}(P(x)u'(x)) = Q(x)u(x) \quad \text{for all } x \in [0, a] \quad \text{with } u(0) = 0 = u(a),$$

where  $a \in (0, 1)$  is given, then  $u$  vanishes on the entire interval  $[0, a]$ . *Hint:* extend  $u$  from  $[0, a]$  to  $[0, 1]$  by setting  $u(x) = 0$  for  $x \in [a, 1]$ . Then integrate by parts to show that  $(u) = 0$ .

8. Consider the variational problem

$$\min \int_0^1 \frac{1}{2}u'(x)^2 + e^{u(x)} dx \quad \text{subject to } u \in H_0^1. \quad (7)$$

- (a) What is the first-order necessary optimality condition (Euler equation) for a local minimizer of this variational problem?  
 (b) Show that if a minimizer of (7) exists, then it is unique.

9. Consider the initial-value problem

$$\dot{\mathbf{x}}(t) = \mathbf{A}(t)\mathbf{x}(t) + \mathbf{u}(t), \quad \mathbf{x}(0) = 0,$$

where  $x : [0, 1] \rightarrow \mathbb{R}^n$  and  $\mathbf{A} \in L^\infty([0, 1]; \mathbb{R}^{n \times n})$ . Obtain a bound for the sup-norm of  $\mathbf{x}$  in terms of the  $L^2$  norm of  $\mathbf{u}$ . *Hint:* Recall Grönwall's inequality

$$x(t) \leq \alpha(t) + L \int_0^t x(s) ds \text{ on } [0, a] \Rightarrow x(t) \leq \alpha(t)e^{Lt} \text{ on } [0, a]$$

where  $\alpha$  is monotone nondecreasing.

10. Consider the following control problem:

$$\min \int_0^1 \frac{1}{2} u^2(x) + y(x) dx \quad \text{subject to } y'(x) = u(x), y(0) = 0, u(x) \geq \ell(x),$$

where  $\ell(x)$  is a given lower bound for the control  $u(x)$  at each  $x \in [0, 1]$ .

- (a) What is the first-order optimality condition (Pontryagin minimum principle) for this problem?
- (b) What is the solution of the control problem?